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Does economic uncertainty predict real activity in real time?☆

Bart Keijsers [a,*], Dick van Dijk [b]

[a] University of Amsterdam, Tinbergen Institute, The Netherlands
[b] Erasmus University Rotterdam, Tinbergen Institute, The Netherlands

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ABSTRACT

We assess the predictive ability of 15 economic uncertainty measures in a real-time out-of-sample forecasting exercise for The Conference Board's coincident economic index and its components (industrial production, employment, personal income, and manufacturing and trade sales). The results show that the measures hold (real-time) predictive power for quantiles in the left tail. Because uncertainty measures are all proxies of an unobserved entity, we combine their information using principal component analysis. A large fraction of the variance of the uncertainty measures can be explained by two factors: a general economic uncertainty factor with a slight tilt toward financial conditions, and a consumer/media confidence index which remains elevated after recessions. Using a predictive regression model with the factors from the set of uncertainty measures yields more consistent gains compared to a model with an individual uncertainty measure. Further, although accurate forecasts are obtained using the National Financial Conditions Index (NFCI), the uncertainty factor models are better when forecasting employment, and in general, the uncertainty factors have predictive content that is complementary to the NFCI.

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1. Introduction

Understanding the fundamental causes of business cycles has intrigued macroeconomists for decades, if not centuries. According to real option theory (Bernanke, 1983; Dixit & Pindyck, 1994), uncertainty is one of the key drivers of such cyclical fluctuations: as uncertainty increases, businesses hold off on investment and consumers postpone large purchases, thus reducing economic activity. Bloom (2009) sparked a new line of research on empirically measuring economic uncertainty and assessing its relationship with real macroeconomic variables such as output and employment; see Bloom (2014) for an overview. This is not a straightforward exercise, because uncertainty is a latent concept and its exact definition can be debated. Not surprisingly then, a variety of measures of economic uncertainty have been proposed over the last decade. Examples include financial volatility (Bloom, 2009), news-based indices (Baker et al., 2016), dispersion in micro-data (Bloom, 2009), and dispersion in forecast errors (Jurado et al., 2015). See Cascaldi-Garcia et al. (2023) for a review on empirical uncertainty measures.

On the introduction of a new measure of economic uncertainty, it is usually added to a vector autoregressive model to assess its impact on macroeconomic variables, typically by means of impulse response functions. The comparison to other measures is usually limited to simple correlations, a visual comparison of extremes, and

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* Correspondence to: Faculty of Economics and Business, University of Amsterdam, Roetersstraat 11, 1018 WB Amsterdam, The Netherlands.
E-mail address: b.j.l.keijsers@uva.nl (B. Keijsers).

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of impulse response functions. All uncertainty measures are proxies of a latent entity, which makes it difficult to assess their quality. This partly explains why a thorough (statistical) comparison of the proposed measures is lacking. Furthermore, evidence of the (dynamic) relationship between uncertainty and economic activity thus far is almost exclusively based on in-sample analysis. It is important for the validity of these findings to test whether the relationship continues to hold out-of-sample. This is important to gain insight into the practical usefulness of the various uncertainty measures. Though in-sample tests have more power (Inoue & Kilian, 2005), a forecasting analysis is relevant for policymakers. Policy decisions often rely on forecasts of economic activity. More accurate forecasts could therefore lead to more informed decision making.

In this paper we address both open issues identified above. First, we collect an extensive set of different uncertainty measures and conduct a factor analysis. This allows us to examine the similarities and differences between the various measures. Furthermore, the resulting factors, essentially combining the information in the different measures, might provide more comprehensive and accurate proxies of (different aspects of) the underlying notion of 'uncertainty.' Second, we conduct a real-time out-of-sample forecasting analysis to assess whether a forecaster is able to take advantage of the implied relationship between uncertainty and economic activity. We specifically compare the predictive ability of uncertainty measures to that of the Chicago Fed's National Financial Conditions Index (NFCI), which captures related information and has been a good predictor of economic activity in recent years (Adams et al., 2021).

For the first part of our analysis, we identify 15 monthly uncertainty measures that comply with a number of restrictions, such as being freely and directly available for a substantial time period. An important additional restriction is that vintages of the uncertainty measures should be available, in case they are subject to revisions. For the uncertainty measures of Jurado et al. (2015), we reconstruct real-time versions thereof, aiming to make our analysis as realistic as possible. The various measures can be categorized into five categories, based on their source: (i) volatility, (ii) cross-sectional dispersion, (iii) news, (iv) surveys, and (v) forecast errors. The collected measures are spread quite evenly across these categories.

The factor analysis shows that there is indeed a fairly strong common component among the uncertainty measures. The first principal component explains about 40% of total variation for the period from 1989–2021. It can be interpreted as general economic uncertainty, because it loads positively on all measures, though slightly more strongly on financial information. Additionally, we identify a second factor, which loads most heavily on news-based and consumer survey-based uncertainty measures. We therefore interpret this factor as media/consumer uncertainty. This second factor remains elevated after recessions, reflecting that media and consumers need more time to become confident about the recovery than reflected by economic fundamentals. Finally, the factors are robust over time. While the Covid-19 period does lead to some differences, the factors remain clearly identified.

For the second part of our analysis, we set up an extensive real-time out-of-sample analysis to forecast The Conference Board's US coincident economic index (CEI), and its components: industrial production, employment, manufacturing and trade sales, and personal income excluding transfer payments. Note that these variables are also taken into account by the NBER's Business Cycle Dating Committee, confirming their importance as measures of real economic activity. In contrast with pseudo-out-of-sample analyses, we use monthly data vintages to take into account that publications of macroeconomic variables are revised multiple times after their initial release. Using these vintages allows us to assess whether a forecaster is able to gain from using the values that are available at that point in time. The importance of employing real-time data in forecasting analyses is discussed by Croushore (2006), among others. The increased interest in computing the downside risk of macroeconomic growth, also known as growth at risk (Prasad et al., 2019), motivated us to forecast quantiles. This provides insight into possible asymmetries in the relationship between uncertainty and macroeconomic activity. Forecasts are produced for the period from 2000 to 2021, based on an expanding window starting in 1990. We consider multiple forecasting horizons, from nowcasting up to 24 months ahead.

We find that the uncertainty measures mostly have predictive ability for the lower quantiles for CEI (and its components') growth rates. This mirrors the asymmetric relationship between GDP growth and the NFCI documented by Adrian et al. (2019), among others. Adams et al. (2021) find that financial conditions improve forecasts of employment, industrial production, and inflation. In comparison with the uncertainty measures, we also find that the NFCI is generally a strong predictor. When forecasting employment, though, a factor model with uncertainty factors performs better at forecasting horizons shorter than 12 months. Interestingly, Bloom (2009) finds that employment responds negatively to uncertainty shocks, and uses this to build a labor-capital model. Moreover, we find that uncertainty factor models hold predictive content complementary to the NFCI for other target variables as well. From the individual uncertainty measures, financial volatility performs best. The performance of individual media and news-based measures is disappointing. The Jurado et al. (2015) measures—from the volatility of forecast errors on a large set of macroeconomic and financial variables—are one of the best performing measures if the final vintage is used. Their predictive ability is substantially worse when using a real-time version that we construct for this exercise. Forecasting accuracy is better and more consistent when using a factor model instead of individual uncertainty measures. So in that sense it is recommended to combine information from multiple uncertainty measures.

Our paper provides three main contributions. First, we add to the literature on the relationship between economic uncertainty and real macroeconomic variables by conducting a real-time quantile forecasting exercise. Second, we show how the different uncertainty measures are related and that they can largely be summarized by two

common factors. Third, we provide further empirical evidence of the relationship between economic uncertainty and the labor market.

There is little research on the forecasting performance of uncertainty measures. Concurrently with this paper, contributions are made by Hengge (2019) and Rogers and Xu (2019). Rogers and Xu (2019) predict GDP growth with a smaller subset of uncertainty measures. Hengge (2019) investigates whether the macro-uncertainty measure of Jurado et al. (2015) predicts the GDP growth rate. We distinguish our analysis by performing a real-time forecasting exercise for different monthly measures of real economic activity, and using a more extensive set of uncertainty measures.

While research on the predictive ability of uncertainty is limited, there is extensive literature on forecasting economic output using measures of financial conditions and risk. We relate to this literature and draw inspiration from a few specific papers. First, Adrian et al. (2019) allow for asymmetry across the density based on quantile forecasts, and find that especially the left tail of GDP growth is affected by financial conditions. Second, most similar to our paper in spirit, Giglio et al. (2016) conduct a quantile forecasting exercise for a set of systemic risk measures. They find that a single common factor improves forecast accuracy, and that predictive power for the mean is limited. The main difference with our paper—other than using uncertainty measures instead of systemic risk measures—is that we conduct a real-time rather than a pseudo-out-of-sample forecasting exercise, taking into account revisions. Systemic risk and financial conditions are close in concept to economic uncertainty. Hence, the findings by Giglio et al. (2016) and Adrian et al. (2019) are consistent with our finding that economic uncertainty is useful for forecasting the lower quantiles of economic output.

The paper is structured as follows. Section 2 describes the uncertainty measures, the selection criteria, and the different categories, followed by the factor analysis in Section 3. Section 4 provides the methodology and implementation details of the real-time forecasting analysis. The forecasting results are presented in Section 5. Section 6 looks into combining the NFCI and uncertainty factors. Section 7 inspects factors based on financial and non-financial uncertainty measures, and Section 8 concludes.

2. Uncertainty measures

Our selection of uncertainty measures is based on a number of criteria. First, we restrict ourselves to US data such that all measures aim to capture the same entity. By far the largest number of measures is available for the US and it makes the results more comparable to the existing literature. Second, to match the frequency of the economic activity variables used in the second part of our analysis, we focus on monthly data. Measures available at a higher frequency are transformed to monthly frequency appropriately. Measures reported at a lower frequency are excluded. They could be included using mixed frequency methods, see e.g. Giannone et al. (2008), but we choose to focus the analysis on a single frequency. Third, the data should be available in real time because we are interested in whether forecasters had been able to take advantage of the information. This excludes measures that are estimated using ex post data, such as forecast error distributions and many other decompositions. Fourth, we require a sufficient time series length such that we have reasonable power for the forecast evaluation. Fifth, on a more practical note, the data should be feasible to collect or compute.

Table A.1 lists the selected uncertainty measures, including a brief description, the source, and the sample size. It is a reasonably sized set of 15 measures from October 1989 to December 2021, and includes most of the popular ones that have been proposed thus far. Notable exclusions are cross-sectional dispersion of firm-level profit growth (Bloom, 2009), total factor productivity growth (Bloom, 2009; Kehrig, 2015), Livingstone Survey GDP forecasts (Bloom, 2009), price changes (Vavra, 2013), and employment growth (Bachmann & Bayer, 2014). Conditional volatility from decomposing financial volatility into risk aversion and uncertainty (Bekaert et al., 2013), shocks from political turmoil, natural disasters or terrorist attacks (Baker & Bloom, 2013), Fama–French factor residual variance (Gilchrist et al., 2014), and fiscal volatility shocks (Fernández-Villaverde et al., 2015) are excluded as well, either because they are only available at a lower frequency, or because they need to be computed ex post. Furthermore, we ignore measures based solely on the Survey of Professional Forecasters (SPF) because these are of quarterly frequency (Lahiri & Sheng, 2010; Rossi et al., 2016).

The descriptions in Table A.1 show that economic uncertainty can be proxied in a variety of ways and from multiple sources. We identify five categories related to how economic uncertainty is measured.[1] First, we consider the volatility of some underlying (often financial) asset. Times of high conditional volatility are assumed to be related to times of high uncertainty. In our set, the underlying assets are stocks (VIX; Bloom, 2009), long-term bonds (MOVE), and the WTI oil price (OVX; Kellogg, 2014).

The second type of uncertainty measure utilizes microdata to estimate cross-sectional dispersion in each time period for a set of individuals, forecasters, or firms. More dispersed outcomes suggest higher economic uncertainty. We consider cross-sectional dispersion in stock returns (CSDR and CSDRsic; Bloom, 2009) and forecast disagreement between respondents from the Philadelphia Fed's Manufacturing Business Outlook Survey (FDISP; Bachmann et al., 2013) and from Consensus Economics GDP growth forecasts (CEgdp; Dovern et al., 2012).

The third source is news, as conveyed via newspaper articles or Bloomberg announcements, among others. In uncertain times, newspapers publish more articles to

[1] Kozeniauskas et al. (2018) and Cascaldi-Garcia et al. (2023) also categorize uncertainty measures. Kozeniauskas et al. (2018) distinguish between macro-uncertainty, micro-uncertainty, and higher-order uncertainty. Cascaldi-Garcia et al. (2023) define text-based, survey-based, econometric-based, and market-based measures.

report on uncertainty and Bloomberg announcements deviate more from expectations. The most prominent measures in this category are the indices from (Baker et al., 2016), based on newspaper article counts, from which we select general economic policy uncertainty (EPU and EPU+) and monetary policy uncertainty (MPU).

Fourth, we consider the outcomes of polls or surveys taken among consumers, professional forecasters, or firms gauging their expectations for the coming period. This is a direct way of measuring the uncertainty perceived by economic agents. For example, the Reuters/University of Michigan Survey of Consumers includes the response "uncertain times" for not buying a vehicle (LLv; Leduc & Liu, 2016) or large household goods (LLh; Fajgelbaum et al., 2017). FDISP, CEgdp, and EPU+ can also be included in this category.

Fifth, uncertainty can be inferred from the volatility of forecast errors. This is to distinguish the uncertainty measure from 'forecastable' time-varying volatility. Jurado et al. (2015) construct uncertainty measures based on this principle. They pool a large set of macroeconomic and financial variables, remove the forecastable part, and compute measures as the stochastic volatility of the forecast errors. The volatility is calculated on subsets of macroeconomic (JLNm), financial (JLNf), and real variables (JLNr).

The dataset is well balanced across the five categories; see Table A.1. There are three measures based on conditional volatility, news, or forecast errors, four measures based on cross-sectional dispersion, and five measures based on survey data.

Only the EPU and JLN measures are subject to revisions. For the EPU measures, vintages are available from 2013M6 (EPU+) or 2019M10 (EPU and MPU). Observations before that period are from the first vintage. More details are provided in Appendix B.1. The JLN measures are not published in real time. These measures are very popular, but quite different from the other measures. Therefore, as an exception, we construct our own monthly vintages using the methodology of Jurado et al. (2015); see Appendix B.2. The vintages are available monthly from 1999M8 onwards.

3. Commonality in uncertainty measures

In the literature, the comparison of uncertainty measures is thus far limited to comparing the pattern of the different time series or computing correlations. Furthermore, it usually involves quite a small set of about four uncertainty measures. We analyze the commonalities for our more extensive set of 15 measures, and assess the underlying factor structure.[2]

Figure A.1 presents the time series of the 15 selected uncertainty measures. They are largely similar in that all measures peak around the time of recessions. Nevertheless, the average correlation between the uncertainty measures is quite modest, at 40.8%. This seems less than expected given that all measures aim to capture US economic uncertainty. The modest average correlation could be due to measurement error, or because the measures account for different aspects of economic uncertainty. The correlation matrix in Figure A.2 indicates multiple blocks of more strongly correlated measures. The average correlation between categories is 34.0%, while the average correlation within a category is 63.0%. In particular, the survey-based measures (excluding FDISP) and the forecasting error-based measures of Jurado et al. (2015) are closely connected, with correlations of 76.8% and 73.8%, respectively. Across categories, measures based on stock market data are similar. For example, the correlation between VIX and JLNf is 83.3%.

The only outlier is FDISP. Other uncertainty measures are correlated by no more than 23% with FDISP, and some are even negatively correlated: −9.9% (with LLv) and −10.2% (with LLh). Figure A.1 confirms the idiosyncratic pattern for FDISP, which could be due to its regional focus or because business surveys capture a unique part of uncertainty.

To assess the commonality between the uncertainty measures more formally, we extract factors using principal components analysis. Table 1 presents the factor loadings and the explained fraction of total variance for the first five principal components. The principal component analysis suggests the presence of two common factors. Together, they explain 63.9% of the total variance and both have a clear interpretation. The first factor represents average (economic) uncertainty, with a slight emphasis on financial uncertainty. The loadings are all positive, and the factor explains almost 50% of the total variance; see Table 1. The factor spikes during recessions and periods of financial turmoil, such as Black Monday in October 1987, the Russian financial crisis of 1998, and the Greek government debt crisis of 2012; see Fig. 1.

The second factor loads most heavily and positively[3] on consumer confidence measures LLv and LLh and news based uncertainty measures EPU and EPU+, see Table 1. This factor can be interpreted as a consumer/media uncertainty factor. Consumers rely on media outlets for economic news, which explains why they are linked. It is interesting to see in Fig. 1 that its value remains relatively high after the recession ended. Apparently, while fundamentals are recovering, the uncertainty among the public remains elevated. This might be because the recovery still has to feed back to consumers, e.g. in the form of new jobs. The unemployment rate typically lags other output variables. This is in line with the jobless recoveries that characterize the periods following the recessions in the 1990s and 2000s (Groshen & Potter, 2003; Jaimovich & Siu, 2020). Further, consumer spending probably lags as well. Consumer savings might be depleted or at least diminished at the end of a downturn, so they probably want to save before spending again. Alternatively, consumers and the media are simply not confident that the recovery

[2] Haddow et al. (2013) and Charles et al. (2018) also use a factor model but on a set of four or six measures only, and their sample excludes the recent Covid-19 period. Section C shows that our factor identification is robust to the Covid-19 period.

[3] The second factor is multiplied by -1. This does not matter for how much of the variance is explained or for forecasting, but makes it more intuitive to explain our interpretation of the factor.

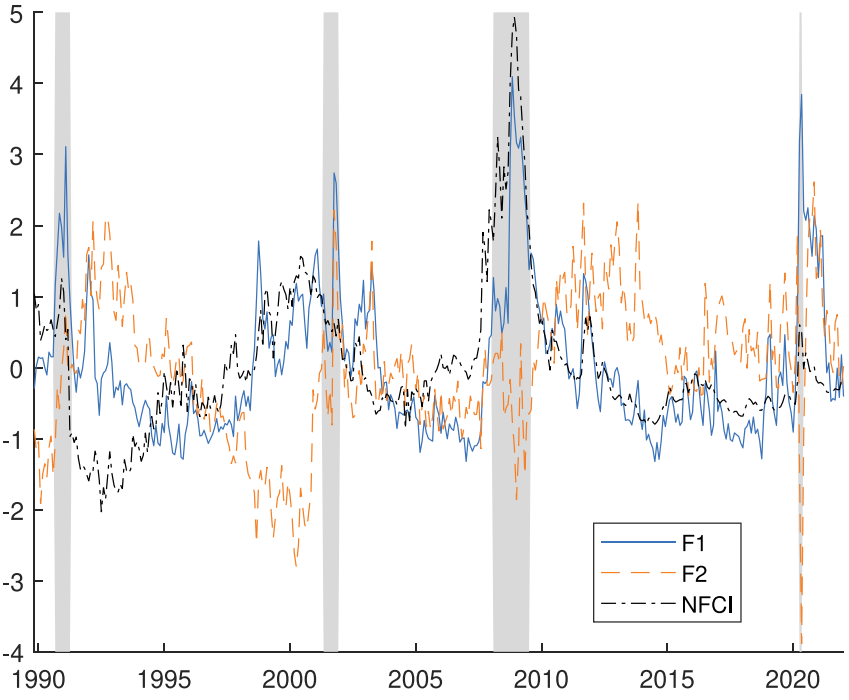


Fig. 1. Uncertainty factors and NFCI.
The figure presents the time series of first releases of the first (solid blue line) and second (dashed orange line) factors from the full sample principal components analysis, and the NFCI (dash-dotted black line). The second factor is rotated (multiplied by −1) for interpretation purposes. The gray bars are recessions as determined by NBER's Business Cycle Dating Committee. All series are standardized.

Table 1
Factor loadings and marginal R^2.

r	1	2	3	4	5
VIX	0.292	−0.206	0.104	0.018	−0.044
MOVE	0.120	−0.384	−0.038	0.533	0.189
OVX	0.245	−0.155	0.236	−0.248	−0.060
CSDR	0.277	−0.292	−0.193	−0.008	−0.282
CSDRsic	0.249	−0.344	−0.228	0.105	−0.224
FDISP	0.052	−0.254	0.540	−0.018	0.653
CEgdp	0.265	0.096	−0.345	0.130	0.318
LLv	0.252	0.376	−0.128	0.223	0.161
LLh	0.267	0.296	−0.200	0.314	0.194
EPU+	0.274	0.360	0.216	0.014	−0.107
EPU	0.265	0.327	0.311	−0.093	−0.230
MPU	0.206	0.006	0.467	0.447	−0.300
JLNm	0.315	0.036	−0.083	−0.364	0.227
JLNf	0.308	−0.204	−0.021	−0.177	−0.118
JLNr	0.334	−0.030	−0.087	−0.324	0.134
R^2	0.478	0.161	0.078	0.070	0.055

The table presents the factor loadings (top panel) and the marginal R^2, the fraction of total variance explained by the rth principal component, for the first five principal components for the sample 1989M10–2021M12, based on the final vintage of the EPU and JLN measures (2022M1). The second factor is rotated (loadings multiplied by −1) for interpretation purposes. See Table A.1 for an explanation of the abbreviations.

has fully started or if it is simply just a coincidental good output number. This is plausible, given that the NBER's Business Cycle Dating Committee usually has a delay of several months in 'officially' calling the end of recessions.

The other factors lack a clear interpretation or explain only a single measure; see Table 1. The third factor explains the variance of mostly FDISP and MPU, but there is no clear link between them. The fourth factor loads heavily on MOVE, and the fifth explains most of FDISP.

3.1. Financial conditions

Financial conditions (NFCI) are closely related to economic uncertainty, or at least capture part of it. The correlations between the uncertainty measures and the NFCI are all positive.[4] As expected, they are quite strongly correlated with the financial measures (65.7% with VIX, 57.1% with CSDR, and 56.4% with JLNf). The strongest pre-Covid-19 correlation is actually with the forecast error-based measures JLNm (79.1%) and JLNr (76.2%). But the response to Covid-19 was so different that the correlations drop to 53.6% and 58.9% when including 2020 and 2021. It is interesting that the correlation with JLNm is high, since JLNm ought to describe macro-uncertainty, while the NFCI captures financial conditions unrelated to other economic conditions. The correlations with consumer survey-based and news-based measures are more modest and in the range of 8.7% to 16.7%.

Fig. 1 shows that the NFCI closely resembles the first uncertainty factor. There are some deviations. For example, the 2001 recession is not captured by the NFCI and the impact of COVID-19 is relatively small. Yet the general

[4] Correlations are based on first-release data, from Amburgey and McCracken (2022).

pattern is very similar. The correlation between the first factor and the NFCI is quite high at 60.7%, and even 80.3% if we exclude 2020 and 2021. This is not surprising given the correlations with the individual uncertainty measures, and given that the first factor loads somewhat more on the uncertainty measures based on financial information. The correlation of the NFCI with the second factor is negative and moderate at −41.4% (−28.9% pre-Covid-19).

4. Methodology

4.1. Coincident variables

Theory suggests a link between economic uncertainty and the business cycle (Bernanke, 1983). For this reason, we consider The Conference Board's Coincident Economic Index (CEI) as our main target variable. In addition, we analyze the predictive ability of uncertainty for the four CEI constituents: industrial production (IP), nonfarm payroll employment (EMP), manufacturing and trade industries sales (MTS), and personal income excluding current transfer receipts (PIX).

In order to assess whether a forecaster is able to improve the accuracy of predictions, real-time data should be used—that is, the vintages with values that were available to the forecaster at the time the forecasts are made. This is relevant because macroeconomic variables are reported with a delay and are subject to revisions. Relying on final vintage data would misrepresent the forecaster's information set; see e.g. Croushore (2006).

Real-time CEI data were obtained from The Conference Board. The four component variables were obtained from the dataset of Chauvet and Piger (2008).[5] The dataset was updated using the Philadelphia Fed's Real-Time Data Set for Macroeconomists (Croushore & Stark, 2001) for industrial production and employment. The most recent vintages for sales and personal income were taken from the St. Louis Fed's ALFRED. For more details, see Appendix D.

The coincident economic index and its constituents are treated as integrated of order 1, and we transform them into annualized growth rates:

$$y_{t+h}^{h,t+h+1} = (1200/h)\log(\mathrm{Y}_{t+h}^{t+h+1}/\mathrm{Y}_{t}^{t+h+1}), \tag{1}$$

where Y_t^v is the original variable at time t from vintage v.

4.2. Quantile forecasts

Based on the link with financial conditions and findings by Giglio et al. (2016) and Adrian et al. (2019), economic uncertainty is expected to mainly affect the left tail of the distribution of the coincident variables. To examine whether the predictive ability of uncertainty indeed varies across the distribution of output growth, we construct quantile forecasts.[6]

Before stating the forecasting model, there are two things to consider. First, following Giglio et al. (2016), we are interested in the quantiles of the shocks to the growth rates of economic activity, rather than the growth rates themselves. These shocks are approximated by residuals from an autoregressive (AR) model with p lags. Second, the aim is to forecast output in real time, emulating reality as closely as possible. Therefore, we use the 'real-time vintage' approach (Clements & Galvão, 2013; Koenig et al., 2003), instead of using end-of-sample data. That is, we use the first release of the data for estimation when available, matching the release maturity of the leading observations on the left- and right-hand side,[7]

$$y_{t+h}^{h,t+h+1} = \beta_0^h + \sum_{j=1}^{p} \beta_j^h y_{t-j+1}^{1,t+1} + u_{t+h}^{h,t+h+1}, \tag{2}$$

for $t = 1, \ldots, T-h$, where $y_t^{h,v}$ is defined in (1). The number of lags $0 \le p \le 6$ is selected using the BIC. The lags on the right-hand side are from the same vintage as the first lag, and can be lightly revised. For example, the second lag will be the second release of that observation. Then, after estimating (2), the shocks are defined as the first-release residuals $\hat{u}_{t+h}^{h,t+h+1}$.

Quantile regression is a semiparametric method dating back to the seminal work by Koenker and Bassett (1978). The estimate of α-quantile $Q_\alpha(y)$ for variable y is the solution to the optimization

$$Q_\alpha(y) = \arg\inf_q \mathrm{E}\left[\rho_\alpha\left(y-q\right)\right], \tag{3}$$

where $\rho_\alpha(x) = \big(\alpha - \mathbf{1}(x \le 0)\big)x$ is the tick loss function, and we specify q as a linear function of (exogenous) regressors. Then, the α-quantile forecasts can be written as

$$Q_\alpha(\hat{u}_{t+h}^{h,t+h+1}|\Omega_{t+1}) = \psi_{\alpha,0}^h + \boldsymbol{\psi}_{\alpha,1}^h \boldsymbol{w}_t^{t+1}, \tag{4}$$

with Ω_{t+1} as the information set at time $t+1$, and $\boldsymbol{w}_t^{t+1}$ as the set of regressors for the uncertainty measures, or factors, or NFCI, at time t from vintage $t+1$. Most regressors are not revised, and can therefore be denoted without a vintage superscript. Financial data are available instantly, but this is not the case for the survey data or the forecast error-based measures. To be consistent and to ensure that the information is available to the forecaster, we use lagged values for all uncertainty measures. Adding more than just the first lag of the regressors $\boldsymbol{w}_t^{t+1}$ did not yield better forecasts.[8]

For values of α, we focus on 0.2, but also analyze the results for 0.1, 0.5 (the median), and 0.8. The parameters $\boldsymbol{\psi}_{\alpha,j}^h$ are estimated using the interior point algorithm. For a review on quantile forecasting, see Komunjer (2013).

[5] To be precise, it is an updated version of the Giusto and Piger (2017) dataset, which updates the Chauvet and Piger (2008) dataset to 2013. Thanks to Jeremy Piger for uploading the raw dataset on his website: https://pages.uoregon.edu/jpiger/research/published-papers/.

[6] In preliminary research, we also checked the ability of uncertainty measures to forecast the mean. The overall results were rather negative. There is little to no forecasting power.

[7] Eq. (2) is slightly different for MTS, because there is a two-month rather than a one-month reporting lag. So to only use data available at the time of forecasting, the vintage is $t+2$ instead of $t+1$. The equation for MTS becomes

$$y_{t+h}^{h,t+h+2} = \beta_0^h + \sum_{j=1}^{p} \beta_j^h y_{t-j+1}^{1,t+2} + u_{t+h}^{h,t+h+1}..$$

[8] In particular, we considered adding up to three lags and selection using the BIC, in line with literature on diffusion forecasting; see e.g. Stock and Watson (2002) or McCracken and Ng (2016).

4.3. Models

As regressors in (4), we consider the following variables. First, the predictive ability of each uncertainty measure is considered individually. That is, $\boldsymbol{w}_t^v = z_{i,t}^v$ is the ith uncertainty measure at time t from vintage v. For the measures other than the EPU and JLN measures, $z_{i,t}^v = z_{i,t}$, since they are not subject to revisions.

Second, we consider the factors extracted from the uncertainty measures using PCA, $\boldsymbol{w}_t^v = \boldsymbol{f}_t^v$, the vector of k uncertainty factors at time t from vintage v. The factors are constructed each month in real time based on the latest vintage data at the point of forecasting. Following the results in Section 3, we consider models with a fixed number of $k = 1$ up to three factors. This allows us to assess the relevance of adding a second or third factor, and to compare against the same model over time. We refrain from estimating the number of factors k because methods of Bai and Ng (2002), Onatski (2010), and Ahn and Horenstein (2013) are for larger panels (larger N) than in our case, and the alternative of a rank test (Cragg & Donald, 1997; Kleibergen & Paap, 2006) is not suitable for covariance matrices; see Donald et al. (2007).

Third, we compare the results from the uncertainty measures and factors to a model with NFCI as the predictor, with $\boldsymbol{w}_t^v = \text{NFCI}_t^v$ as the end-of-month NFCI at time t from vintage v. We use the unofficial real-time version constructed by Amburgey and McCracken (2022), which coincides with the official Fed vintages from May 2011 onwards.[9]

Finally, as a benchmark, the performance of the models is compared to the historical quantile estimate $\hat{q}_{\alpha,t}$, the empirical quantile based on data up to and including time t.[10]

4.3.1. Sample

In the forecasting exercise, we recursively estimate all models. That is, at each time t, we first estimate the factors and models using data from 1989M12 to time $t-h$. Earlier (initial) observations are included if the lag order is larger than one. We start in 1989M12 because that is the first period where at least three months of data are available for all variables. Recursive estimation is in line with other diffusion forecasting literature, see e.g. Stock and Watson (2002) and McCracken and Ng (2016). Using all available information improves the convergence of the factor estimates. Moreover, results using a rolling window did not indicate the presence of a structural break, while the forecasting results deteriorate in some cases.

Second, the parameter estimates and time t observations are used to construct the forecast for the $t+h$ value $y_{t+h}^{h,t+h+1}$. We imagine a forecaster who starts forecasting in January 2000. The first forecast is made for period 1999M12$+h$, and the final one for 2021M12, for horizons h of 1 (nowcast), 3, 6, 12, and 24 months. This yields a sample of 120 initial in-sample and 265$-h$ out-of-sample observations.

4.4. Evaluation

A relevant question with real-time data is which values to use as 'actuals' to evaluate the forecasts. Preferably, these are the true values that are no longer revised. This is impossible, however, because of benchmark revisions. For example, due to a change of the index year, annual updating follows the consensus numbers and redefinitions or measurement changes. Broadly, there are three alternatives.

One option is to use the xth release observations $y_{t+h}^{h,t+h+x}$ for some $x \geq 1$. Many empirical studies use xth release data to evaluate their forecasts; see e.g. Romer and Romer (2000), Groen et al. (2013) and D'Agostino et al. (2013). Selecting x requires some knowledge of the revision process. For quarterly data, the second revision (third release) is often used because this is usually the 'final' revision from the statistical agency.

A second option is to use the final vintage observations $y_{t+h}^{h,T+1}$. The final vintage is the most recent publication of the numbers. For example (Koenig et al., 2003) and Clements and Galvão (2013) use the vintage published about a year and a half after the end of their sample. An advantage is that it incorporates the latest available information and is currently closest to the true values as a single time series.

The third option is to use the pre-benchmark observations $y_{t+h}^{h,\text{PBM}}$ as actuals. Pre-benchmark values are the final observation before the first benchmark revision after a first value for a given date has been reported. We choose to use this third option. In contrast to regular non-benchmark revisions, benchmark revisions can and should not be predictable to the forecaster (Croushore, 2006). An additional argument to use pre-benchmark values as actuals is that the actuals should represent the forecasters' target rather than be the closest possible to the current truth. Both xth release and final vintage observations are subject to benchmark revisions. Some pre-benchmark observations are still subject to regular revisions though.

To evaluate the quantile forecasts, the pre-benchmark residuals that we use as actuals are defined as

$$\hat{u}_{t+h}^{h,PBM} = y_{t+h}^{h,PBM} - \left(\hat{\beta}_0^h + \sum_{j=1}^{p} \hat{\beta}_j^h y_{t-j+1}^{1,t+1} \right), \tag{5}$$

where p is same value as in Eq. (2) and $\hat{\beta}_j^h$ is estimated from the model in Eq. (2). Because the estimated model using first vintage data is an efficient estimate of the actual mean (Koenig et al., 2003), we can use it to 'demean' the actuals too to get the actual shocks. So the mean is consistent across estimation and evaluation.

The relative forecast accuracy of the quantile forecasts is evaluated by comparing the mean tick loss (MTL),

[9] The unofficial NFCI vintages are available on McCracken's website: https://research.stlouisfed.org/econ/mccracken/fred-databases/.

[10] As another benchmark, we considered a factor model with factors from the FRED-MD dataset (McCracken & Ng, 2016) as predictors, where real-time data are available from the 1999M08 vintage (but published in real time from the 2015M01 vintage) and the number of factors is selected using the BIC, with a maximum of eight factors. The FRED-MD factors' predictions are a bit disappointing at the lower quantiles and oftentimes do not even beat the historical quantile. The results are included in Appendix G.

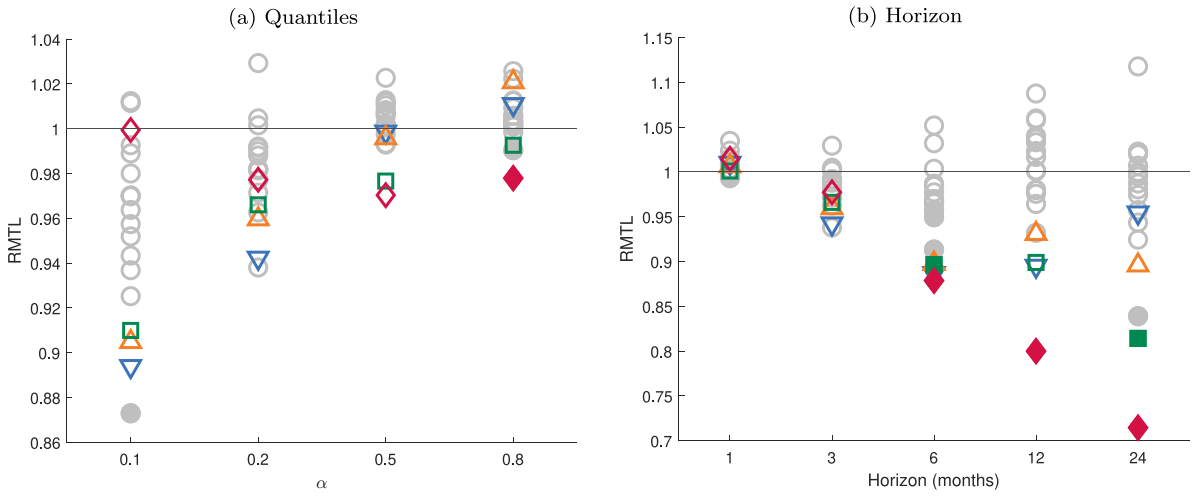


Fig. 2. RMTL from forecasting coincident economic index.
The figures present the relative mean tick loss (RMTL), with the historical quantile as benchmark, from forecasting the coincident economic index for multiple quantiles and forecast horizons. The forecasting horizon for the different quantiles is three months. The quantile for the different horizons is $\alpha = 0.2$. Gray circles are models with a single uncertainty measure. Blue down-pointing triangles, orange up-pointing triangles, and green squares denote models with one, two, and three uncertainty factors, respectively. Red diamonds are models with the NFCI. Filled symbols indicate significance of the one-sided DM test against the historical quantile at the 5% significance level. The RMTL values are presented in tables in Appendix G.

which is a proper scoring rule for quantiles (Gneiting & Raftery, 2007). To facilitate a comparison, we compute the MTL relative to the historical quantile's mean tick loss (RMTL). If the RMTL is below 1 for a model, it has a smaller loss and thus yields on average more accurate quantile forecasts than the historical quantile benchmark. Statistical significance is tested using one-sided (Diebold & Mariano, 1995) tests, where we test the null of equal predictive accuracy, versus the alternative of smaller loss compared to the benchmark model. We follow the arguments by Faust and Wright (2013) by relying on the Monte Carlo evidence presented by Clark and McCracken (2013). Their simulation study shows that the Diebold–Mariano test statistic with standard normal critical values and the corrections by Harvey et al. (1997) yields satisfactory size, even for nested models.

5. Forecasting results

We first examine the performance in forecasting the CEI. The RMTLs are plotted in Fig. 2 for different horizons and quantiles.[11]

5.1. Uncertainty measures

Fig. 2(a) immediately shows that the uncertainty measures yield better forecasts at lower quantiles than at higher quantiles. The average RMTL is 0.963 for quantile 0.1. All but two measures yield an RMTL below 1, while it is on average 1.005 at quantile 0.5 with only three out of 15 measures outperforming the benchmark.

A few specific uncertainty measures stand out in terms of predictive power: OVX and VIX. First, the model with OVX as the predictor yields the smallest tick loss for the uncertainty measures at the short horizon (up to three months) for the quantiles 0.1 and 0.2. At the three-month horizon, the model with OVX significantly outperforms the historical quantile by 12.7% for quantile 0.1 and by 6.2% for quantile 0.2. Second, the VIX yields good forecasts at medium horizons, with p-values under 10% between three- and 12-month horizons at quantile 0.2. The gains in terms of tick loss are between 3% (three-month horizon) and 9% (six-month horizon) compared to the historical quantile. Models with other individual uncertainty measures do not perform as well consistently. Uncertainty measures that perform surprisingly poorly are EPU and EPU+. Despite their popularity, they do not yield better forecasts than the benchmark in most of the cases when predicting the CEI.

Comparing to the NFCI, the uncertainty measures yield comparable tick losses at the short horizon. From the six-month horizon, however, the NFCI is better than all individual uncertainty measures. The gap is largest at the 12- and 24-month horizons with an average increase in tick loss of 23% and 30% compared to the NFCI. Moreover, the NFCI is significantly more accurate than the benchmark at the six- to 24-month forecast horizons.

5.2. Uncertainty factors

Turning to the uncertainty factor models, Fig. 2 shows that they generally perform on par with or better than the best individual uncertainty measures. In particular at the medium horizon, the uncertainty factor models perform well, with gains in tick loss of 11.3% and 10.5% compared to the benchmark at the six- and 12-month horizons for the 0.2 quantile for the one-factor model. The

[11] See Appendix F for an evaluation of the absolute performance of the quantile forecasts by means of Engle and Manganelli's (2004) DQ tests to verify that the coverage is in line with the expected level, and whether the hits—observations smaller than the predicted quantile—are not predictable. The results indicate reasonably good coverage.

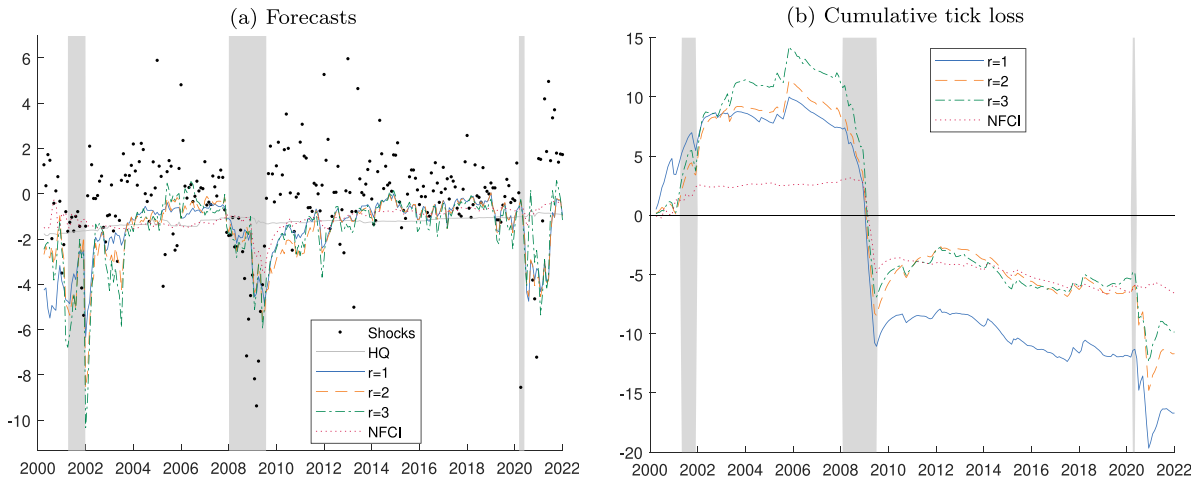


Fig. 3. Tick loss for coincident index over time.
The left figure presents the realized values and the quantile forecasts for the coincident economic index at quantile 0.2 for a three-month horizon from the historical quantile (solid gray line), factor models (with r factors) and the model with the NFCI. The right figure presents the cumulative tick loss over the evaluation period minus the cumulative tick loss from the historical quantile.

forecasts are (close to) significantly better than the benchmark with p-values of 3.3% (six-month horizon) and 6.2% (12-month horizon). Importantly, the gains when using a factor are more consistent across horizons and (lower) quantiles compared to individual uncertainty measures.[12] Even though the factors yield more consistent results across the horizons, the NFCI is still more accurate at the longer horizons (from six months onwards).

While the number of factors needed to explain the commonality in the uncertainty measures seems to be two, this is not necessarily the number that yields the best forecasts. In the case of the CEI, the second factor does not hold much additional predictive power over the first factor. In fact, the two-factor model often leads to a slightly higher tick loss, although the differences are small.

5.3. Tick loss over time and impact of COVID-19

Perhaps unsurprisingly, most of the gains in forecast accuracy compared to the benchmark are achieved during recessionary periods, and specifically during the financial crisis in 2008; see Fig. 3(a). This holds for models with individual uncertainty measures, as well as the factor models and NFCI. The historical quantile does not capture the downturns. The quantile is overestimated, and due to the tick loss function, this leads to a relatively large loss. Outside of recessions, the historical quantile is hardly outperformed by the models under consideration.

The Covid-19 period differs from the other recessions in that the underlying reason is not economic and therefore hard to predict using economic uncertainty and financial conditions.[13] Indeed, including 2020 and 2021 substantially increases the MTL for all models. At the three-month horizon for quantile 0.2, the MTL of the historical quantile almost doubles from 0.566 to 1.105. Other models are impacted even more, as the relative tick loss also worsens compared to the pre-Covid period. Again for the three-month horizon and the 0.2 quantile, the RMTL increases for 10 out of 15 uncertainty measures when including 2020 and 2021 in the sample period. Still, they at least somewhat capture the downturn as the average increase in RMTL over all uncertainty measures is only 2.2 percentage points. The same holds for the uncertainty factor models, where the RMTL increases with 0.5 to 3.0 percentage points. Also the NFCI does not capture the post-Covid-19 dynamics better than the benchmark, with an increase of 2.5 percentage points in RMTL. Ultimately though, the ordering of the models' performance and the significance levels of the forecasting performance are not affected much.

5.4. Forecasting the components of the coincident index

The forecasting results for the components of the coincident index (industrial production, employment, manufacturing and trade sales, and personal income) are largely consistent with those for the CEI. The most important and consistent finding is that uncertainty measures and factors outperform the historical quantile when forecasting the left tail; see Fig. 4 for an example at the three-month horizon.

The forecasting results are most convincing for employment. At the three-month horizon for quantile 0.2, all

[12] In Section E.1, as an alternative aggregation method, we inspect the forecasting performance when averaging uncertainty measures within the categories from Section 2. In short, the results confirm the findings from the individual uncertainty measures: conditional volatility, cross-sectional dispersion, and forecast errors yield the smallest RMTL. Which category holds the most predictive information depends on the horizon and target variable. Forecasts from the factor models are more accurate or on par with the category averages, and are more consistent across horizons and target variables.

[13] For discussions on how to treat the Covid-19 period when modeling or forecasting macroeconomic variables, see e.g. Carriero et al. (2022), Lenza and Primiceri (2022), and Schorfheide and Song (2021).

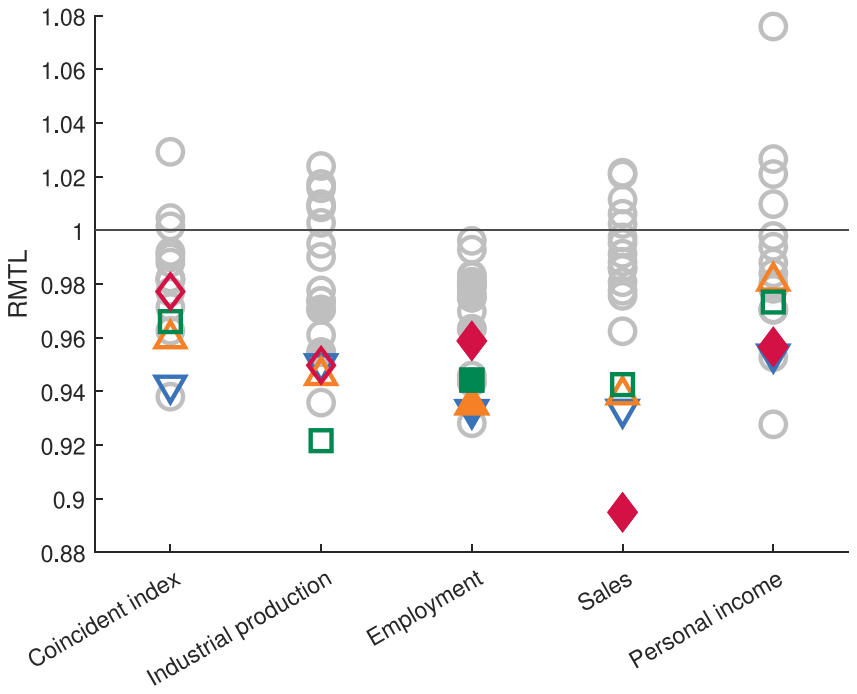


Fig. 4. Tick loss across target variables.
The figure presents the relative mean tick loss (RMTL), with the historical quantile as benchmark, from forecasting the coincident economic index and its component: industrial production, employment, manufacturing and trade sales, and personal income. The forecasting horizon is three months. The quantile is $\alpha = 0.2$. Gray circles denote models with a single uncertainty measure. Blue down-pointing triangles, orange up-pointing triangles, and green squares are models with one, two, and three uncertainty factors, respectively. Red diamonds are models with the NFCI. Filled symbols indicate significance of the one-sided DM test against the historical quantile at the 5% significance level. The RMTL values are presented in Appendix G in tables.

uncertainty measures beat the historical quantile, and the uncertainty factors render the best performance with improvements of 5.6% to 6.7% in mean tick loss compared to the benchmark. Moreover, the difference in tick loss is significant for all factor models at the 5% significance level. The uncertainty measures that yield significant gains are mostly those based on financial information: VIX, MOVE, CSDR(sic), and MPU. This is in line with Bloom (2009), who considers the effect of uncertainty on the labor market using the VIX (and CSDR(sic)).

When forecasting employment, the uncertainty factor models even edge out the NFCI. At the three-month horizon for the 0.2 quantile, the uncertainty factor models have a 1.5% (three factors) to 2.7% (one factor) lower tick loss compared to the NFCI—although, according to the DM tests there is no significant improvement. There is some weak evidence at the 0.1 quantile with p-values of 5.8% (one-month horizon) and 7.2% (three-month horizon) that the factor models yields statistically better forecasts than the NFCI.

The gains are similar at the 0.1 and 0.2 quantile forecasts of employment (see Fig. 5(a)) while there is little predictive power at the middle and higher quantiles. At the middle and higher quantile forecasts of employment, there is little predictive power; see Fig. 5(a). Further, Fig. 5(b) shows that the factor models perform best at the short to medium forecasting horizon up to 12 months.

5.5. Relevance of real-time data

What are the consequences of ignoring revisions in the uncertainty measures and using the latest available observations? We address this question by repeating the forecasting exercise using the final vintage of the EPU and JLN measures (data up to 2021M12, as published on Ludvigson's website).[14] When using these observations instead of the real-time data, the RMTL for the models with the EPU measures still perform poorly. By contrast, the models with the final vintage JLN measures improve drastically, in line with findings by Rogers and Xu (2019). The RMTL decreases across all target variables from a three-month or longer horizon. The relative gains are larger at longer horizons, with gains in the 3.4 to 16.3 percentage point range at the six- and 12-month horizons for JLNm and JLNf. The most dramatic decrease in RMTL is 34.8 percentage points when forecasting employment at the 24-month horizon with the last vintage JLNr instead of the real-time version. All JLN measures would be among the best individual uncertainty measures.

Using final vintage data also improves the forecasting performance of the uncertainty factor models, in particular for employment. The RMTL for the two-factor model decreases by 1.4 to 5.9 percentage points from the three- to the 24-month horizons. When using real-time data, the

[14] We retain the real-time data for the target variables.

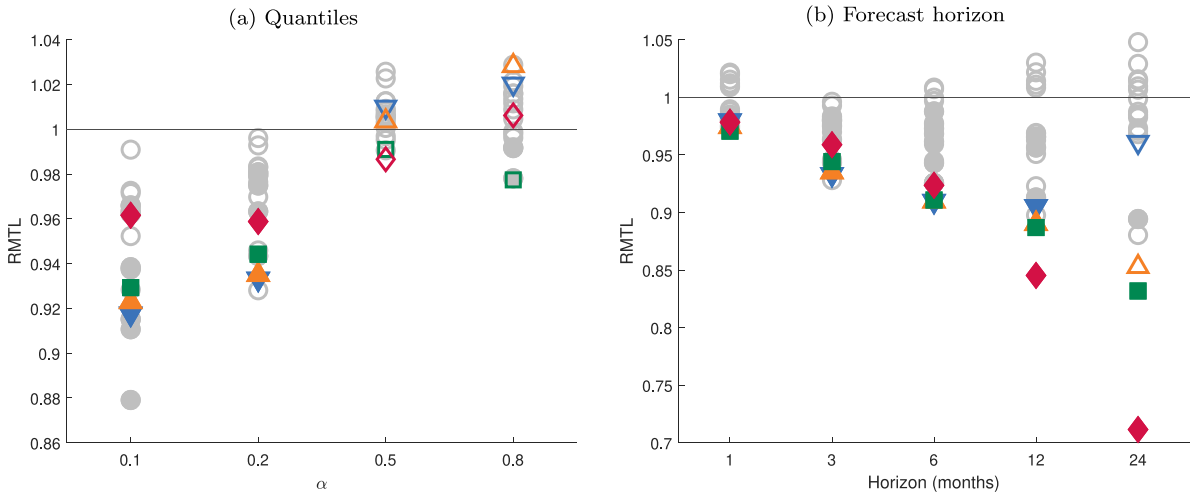


Fig. 5. Tick loss in forecasting employment.
The figures present the relative mean tick loss (RMTL), with the historical quantile as benchmark, from forecasting non-farm payroll employment for multiple quantiles and forecast horizons. The forecasting horizon for the different quantiles is three months. The quantile for the different horizons is $\alpha = 0.2$. Gray circles denote models with a single uncertainty measure. Blue down-pointing triangles, orange up-pointing triangles, and green squares are models with one, two, and three uncertainty factors, respectively. Red diamonds are models with the NFCI. Filled symbols indicate significance of the one-sided DM test against the historical quantile at the 5% significance level. The RMTL values are presented in Appendix G in tables.

results are more modest. A forecaster should therefore be cautious in interpreting forecasting results based on final vintage data.

6. Economic uncertainty and financial conditions

NFCI outperforms the uncertainty measures and factor models in many cases; see Figs. 2 and 4. The question is whether there is relevant information in the uncertainty measures that is not present in the NFCI. Table 2 presents the results from the Giacomini and Komunjer (2005) quantile encompassing test.[15] The table shows whether none, both, or only one of the forecasts should be used. In the latter case, that model's forecast encompasses the excluded one. Where it says both, a combination of the quantile forecasts is better than only one of the two.

Two conclusions follow from the encompassing tests. First, at the 0.2 quantile it can be valuable to add uncertainty factors when forecasting at short to medium horizons up to 12 months. For employment at the three-month horizon, there is some evidence that the uncertainty factor model encompasses the NFCI. Second, the NFCI encompasses the uncertainty factor model in many cases, especially at horizons of six and 12 months. This is somewhat contrary to Hengge (2019), who finds, using the predictive score, that the predictive power of JLNm is not impacted by including the NFCI. Diks et al. (2011) show that using a different scoring rule can substantially affect the conclusions though. Third, the uncertainty factor models do comparatively better further into the tail. At the 0.1 quantile, it is more clear that the uncertainty factor model tends to encompass the NFCI at shorter horizons up to six months, and vice versa at longer horizons.

To verify the encompassing results, we combine the quantile forecasts from the model with the NFCI and the uncertainty factor model in two ways. First, we include both forecasts as regressors in a quantile regression (also including an intercept) and estimate the weights recursively—the evaluation sample starts at 2002M12+h—to allow for a burn-in period for the estimated combination weights. This specification corresponds to the model in the encompassing test. Second, we also consider an equally weighted combination, which is known to work well in general (see e.g. Timmermann, 2006).

The encompassing results are largely corroborated by the performance of the forecast combinations; see Fig. 6. As expected from Table 2, the model with only the NFCI is the best model at the longer horizons of 12 and 24 months, except for employment at the 24-month horizon.

At shorter horizons up to six months, combining forecasts often yields a smaller RMTL. This is almost only when considering an equally weighted combination of the uncertainty factor model and the NFCI model forecasts. The forecast combination based on quantile regression is actually the worst model in many cases. On average, the RMTL at the three- and six-month horizons is 1.4 and 6.3 percentage points smaller when using equal weights instead of estimated weights. This is in line with the known robustness of equally weighted forecast combinations (Timmermann, 2006).

So, economic uncertainty seems to hold some relevant predictive information that is not captured by financial conditions.

[15] The unconditional version. To allow for misspecification, we implement the test with an intercept and allow the competing quantile's coefficient to differ from 1. That is, the combined quantiles are $\hat{q}_{ct} = \theta_0 + \theta_1 \hat{q}_{1t} + \theta_2 \hat{q}_{2t}$, where $\hat{q}_{1t}$ and $\hat{q}_{2t}$ are the quantile predictions from the competing models, and we test whether $\theta_1 = 0$ and/or $\theta_2 = 0$. Standard errors are computed using a Newey–West estimator with $h-1$ kernel width, following Giacomini and Komunjer (2005).

Table 2
Preferred model from encompassing test.

Target variable	Horizon (months)				
	1	3	6	12	24
	$\alpha = 0.1$				
Coincident index	Uncertainty	Uncertainty	Both	None	None
Industrial production	Uncertainty	Both	NFCI	NFCI	NFCI
Employment	Uncertainty	Both	Uncertainty	Both	None
Manufacturing and trade sales	None	NFCI	NFCI	Both	Both
Personal income	Uncertainty	None	Uncertainty	Both	Both
	$\alpha = 0.2$				
Coincident index	Both	Uncertainty	Both	NFCI	None
Industrial production	NFCI	Uncertainty	Both	NFCI	None
Employment	Both	Uncertainty	Both	NFCI	NFCI
Manufacturing and trade sales	NFCI	NFCI	NFCI	NFCI	Both
Personal income	Uncertainty	NFCI	NFCI	NFCI	Both

The table presents what model is preferred, based on the (Giacomini & Komunjer, 2005) encompassing test at a 5% significance level, for forecasts of quantiles $\alpha = 0.1$ and $\alpha = 0.2$. The competing models are the factor model with the first two uncertainty factors and the model with NFCI.

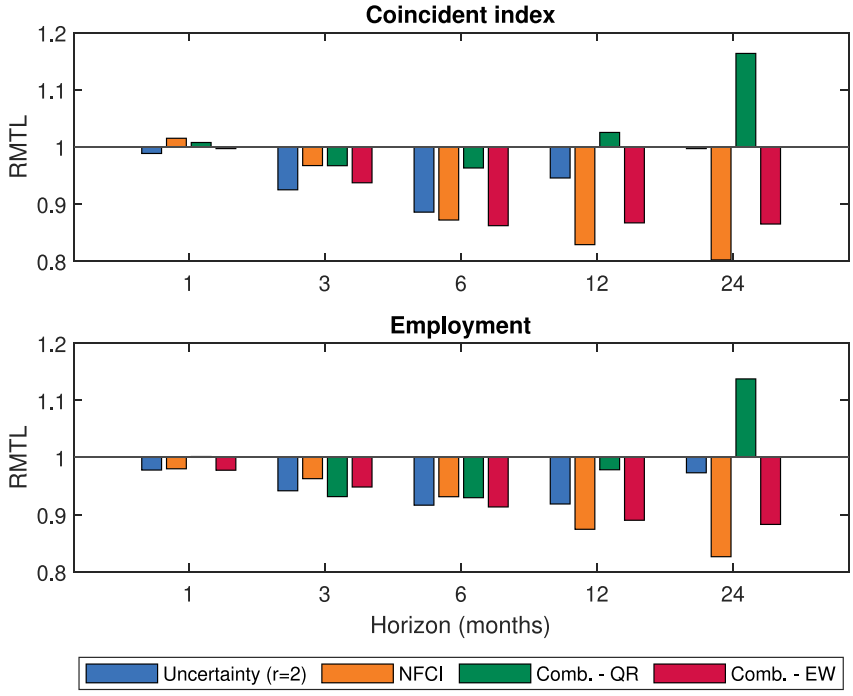


Fig. 6. RMTL of forecast combinations.
The figure presents the relative mean tick loss (RMTL), with the historical quantile as benchmark, from forecasting the coincident economic index and employment. The quantile is $\alpha = 0.2$. Models include the uncertainty factor model with $r = 2$ factors (blue bars), the model with the NFCI as regressor (orange bars), and two forecast combinations that combine the forecasts from the model with two uncertainty factors and the model with the NFCI, where the quantiles forecasts are weighted by recursively estimating a quantile regression (QR, green bars) or equally weighted (EW, red bars). The evaluation period is from 2002M12+h–2021M12, to allow for a two-year burn-in period for the weights in the forecasting combination. The results from forecasting industrial production, sales, and personal income are in Figure G.1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

7. Financial- and non-financial-based uncertainty

The comparison with the NFCI raises the question of what information is relevant for forecasting the coincident economic index and its components, and whether this is related to financial conditions or macroeconomic uncertainty. Caldara et al. (2016) and Ludvigson et al. (2021) also differentiate between macroeconomic and financial uncertainty. Caldara et al. (2016) report a negative effect on economic activity from both sources. Ludvigson et al. (2021) find that high financial uncertainty is a cause of real activity shocks, while increases in macroeconomic uncertainty are a response to it.

Therefore, we split the uncertainty measures into two groups: those based on financial information (VIX, MOVE,

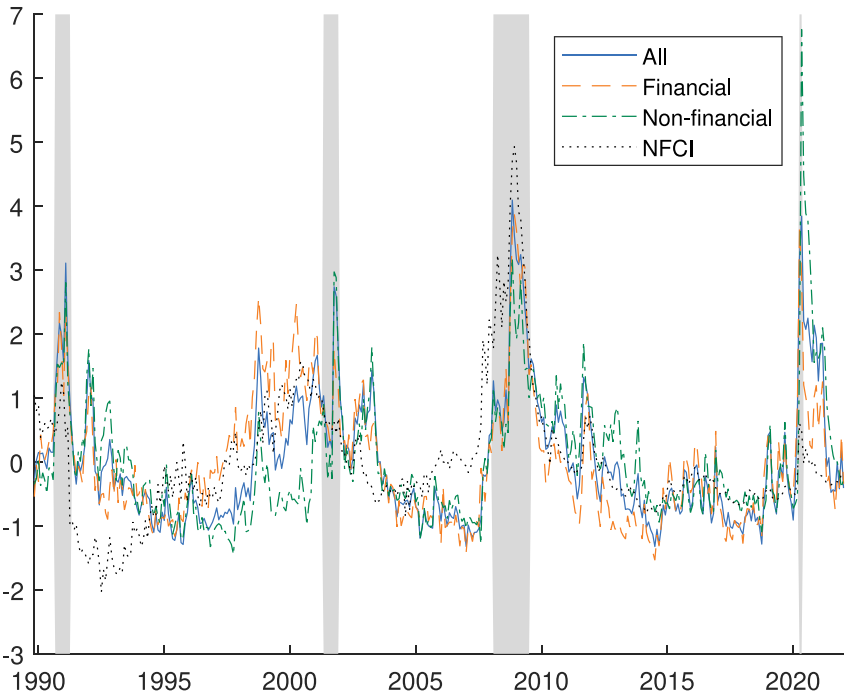


Fig. 7. First factor from subsets of measures.
The figure presents the time series of first releases of the first factor from the principal components analysis based on the full set of uncertainty measures (solid blue line), the subset of financial-based uncertainty measures (dashed orange line), and the subset of non-financial-based uncertainty measures (dash-dotted green line) and the NFCI (dotted black line). The gray bars are recessions as determined by NBER's Business Cycle Dating Committee. All series are standardized.

OVX, CSDR, CSDRsic, and JLNf) and those based on non-financial information. Then, we compute the first factor from the principal components analysis for each group separately.

At first sight, the first factors from both subsets of uncertainty measures seem to be very similar (see Fig. 7) with correlations of 87.7% (financial) and 86.0% (non-financial) with the first factor obtained from the full set. The correlation between the factors of the subsets is only 55.1% though. Where the factors deviate most is during the period from 1996–2001—likely due to the Asian and Russian financial crises and the dot-com bubble. Also, the factor from financial-based uncertainty measures attains a higher value during the financial crisis in 2007–2008, and responds less strongly to the Covid-19 crisis compared to the factor from non-financial uncertainty measures. As expected, this pattern for the factor from financial-based uncertainty measures is more in line with the NFCI and is also seen in the correlations: 66.2%, which is higher than the factor based on the full set of measures (60.7%) or the non-financial-based measures (34.8%).

The second factor from the full set is captured less strongly by the factors from the subsets. The largest (absolute) correlation is 73.5%, with the second factor from the non-financial subset, and 40.1% for the financial subset.

Since the NFCI outperforms the uncertainty factor models in many cases, we expect that the factors based on the financial-based uncertainty measures outperform the other factors models as well. Table 3 shows that this is partly true. Using the first financial-based uncertainty factor leads to a smaller MTL compared to using the first factor based on non-financial information. However, neither outperforms the factor model based on the full set of uncertainty measures. Except for industrial production, where the single factor model based on the financial uncertainty measures performs close to the model with NFCI. At horizons of 12 months and more, though, the NFCI yields smaller RMTL. This could be because the NFCI is constructed from indicators of three categories: risk, credit, and leverage indicators. By contrast, the financial uncertainty measures are associated with the risk category only.

Interestingly, the second factor contains important predictive power when considering the non-financial set, in particular when forecasting employment. The second factor loads heavily on FDISP. Individually, FDISP does not decrease the MTL by much compared to the benchmark, up to 3.1% for a 12-month horizon or less. Though the gains are significant at all but the three-month horizon. The non-financial factor models yield good forecasts for employment also in the first ten years of the sample—from January 2001 to December 2009, so including the financial crisis. The non-financial factor models yield the smallest tick loss, with an RMTL of 0.820 (one factor) and 0.815 (two factors) at the three-month horizon and 0.2 quantile. So it seems that most relevant information comes from the financial uncertainty measures, but not all predictive power.

Table 3
RMTL for factor models on subsets.

Horizon (months)	1	3	6	12	24
Panel A: Coincident economic index					
Financial information-based measures					
$r = 1$	0.999	0.960**	0.906***	0.924**	0.931
$r = 2$	1.001	0.960**	0.923**	0.951*	0.933
Non-financial information-based measures					
$r = 1$	1.022	0.966	0.962	0.990	0.992
$r = 2$	1.021	0.967	0.921	0.957	0.911
Panel B: Nonfarm payroll employment					
Financial information-based measures					
$r = 1$	0.972***	0.956***	0.930***	0.917***	0.965
$r = 2$	0.968***	0.954***	0.933***	0.930**	0.989
Non-financial information-based measures					
$r = 1$	1.000	0.959	0.947	0.969	0.992
$r = 2$	0.983*	0.930**	0.908***	0.867**	0.797**

The table presents the relative mean tick loss for various forecasting horizons and target variables, for the full sample and quantile $\alpha = 0.2$. The results from forecasting industrial production, sales and personal income are in Table G.4.
* Denote significance of a one-sided Diebold–Mariano test at the 10%.
** Denote significance of a one-sided Diebold–Mariano test at the 5%.
*** Denote significance of a one-sided Diebold–Mariano test at the 1%.

8. Conclusion

Many economic uncertainty measures have been proposed over the last 15 years. We showed that they share a factor structure. The first common component explains over 40% of the total variation. The second factor can be interpreted as media/consumer uncertainty, which tends to remain high after officially leaving recessions.

The results of our real-time forecasting analysis showed that there is a non-linear relation—between the uncertainty measures and factors, and the coincident economic index—that can be utilized to forecast the lower quantiles of the index. The VIX and OVX are recommended individual measures, but using the factors is preferred for more consistent gains. The predictive content of economic uncertainty is relevant for professional forecasters and policymakers to keep an eye on, in particular when interested in the labor market. Moreover, at shorter horizons, the uncertainty measures seem to hold predictive content in addition to financial conditions, as captured by the NFCI.

CRediT authorship contribution statement

Bart Keijsers: Conceptualization, Data curation, Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Dick van Dijk:** Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

A few restrictions apply to the data. The CEI vintages from The Conference Board, the MOVE index from ICE, and the source data for CEgdp from Consensus Economics are proprietary and cannot be shared. The other target variables (see Section D) and uncertainty measures (see Table A.1) are freely available. The components to compute the JLN vintages (see Section B.2) can also be downloaded without restriction. Unofficial real-time NFCI vintages are available from Michael McCracken's website (https://research.stlouisfed.org/econ/mccracken/fred-databases/).
The data are prepared in Matlab and Python. Real-time JLN vintages are computed in Matlab and R, based on code from Sydney Ludvigson's website (https://www.sydneyludvigson.com/macro-and-financial-uncertainty-indexes). The analysis is implemented in Matlab, where quantile regressions are estimated using code from Roger Koenker's website (http://www.econ.uiuc.edu/~roger/research/rq/rq.html), and the code for LASSO quantile regression is from Alexandre Belloni's website (https://people.duke.edu/~abn5/belloni-software.html).

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2024.06.008.

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